Linear Algebra Proofs Compendium (Midterm II Master Reference)

Comprehensive Exam Proofs: Vectors in $\mathbb{R}^n$ through Linear Transformations

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Part I: Vectors, Projections, Inner Products & Cauchy-Schwarz ($\mathbb{R}^n$)

Theorem 3.3.3: The Projection Theorem

Statement

> If $\mathbf{u}$ and $\mathbf{a}$ are vectors in $\mathbb{R}^n$ and $\mathbf{a} \neq \mathbf{0}$, then $\mathbf{u}$ can be expressed in **exactly one way** as:
>
> $$\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$$
>
> where $\mathbf{w}_1$ is a scalar multiple of $\mathbf{a}$ (the orthogonal projection $\text{proj}_{\mathbf{a}}\mathbf{u}$) and $\mathbf{w}_2$ is orthogonal to $\mathbf{a}$:
>
> $$\mathbf{w}_1 = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a}, \qquad \mathbf{w}_2 = \mathbf{u} - \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a}$$

Proof

1. Existence (Constructing $\mathbf{w}_1$ and $\mathbf{w}_2$)

Let $\mathbf{w}_1 = k\mathbf{a}$ for some scalar $k \in \mathbb{R}$, and define $\mathbf{w}_2 = \mathbf{u} - k\mathbf{a}$. Clearly, $\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{u}$ and $\mathbf{w}_1$ is a scalar multiple of $\mathbf{a}$.

We require $\mathbf{w}_2$ to be orthogonal to $\mathbf{a}$:

$$\mathbf{w}_2 \cdot \mathbf{a} = 0$$

Substitute $\mathbf{w}_2 = \mathbf{u} - k\mathbf{a}$:

$$(\mathbf{u} - k\mathbf{a}) \cdot \mathbf{a} = 0$$

Using the distributive property of the dot product:

$$\mathbf{u} \cdot \mathbf{a} - k(\mathbf{a} \cdot \mathbf{a}) = 0$$

$$\mathbf{u} \cdot \mathbf{a} = k\|\mathbf{a}\|^2$$

Since $\mathbf{a} \neq \mathbf{0}$, we have $\|\mathbf{a}\|^2 \neq 0$, so we can divide by $\|\mathbf{a}\|^2$:

$$k = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}$$

Thus, setting $\mathbf{w}_1 = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a}$ and $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1$ gives a valid decomposition.

2. Uniqueness

Suppose $\mathbf{u}$ can also be written as $\mathbf{u} = \mathbf{w}_1' + \mathbf{w}_2'$ where $\mathbf{w}_1' = k'\mathbf{a}$ and $\mathbf{w}_2' \cdot \mathbf{a} = 0$. Since both decompositions sum to $\mathbf{u}$:

$$\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{w}_1' + \mathbf{w}_2'$$

$$k\mathbf{a} + \mathbf{w}_2 = k'\mathbf{a} + \mathbf{w}_2' \implies (k - k')\mathbf{a} = \mathbf{w}_2' - \mathbf{w}_2$$

Take the dot product of both sides with $\mathbf{a}$:

$$(k - k')(\mathbf{a} \cdot \mathbf{a}) = (\mathbf{w}_2' - \mathbf{w}_2) \cdot \mathbf{a} = \mathbf{w}_2' \cdot \mathbf{a} - \mathbf{w}_2 \cdot \mathbf{a} = 0 - 0 = 0$$

Since $\mathbf{a} \neq \mathbf{0}$, $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \neq 0$, which forces:

$$k - k' = 0 \implies k = k'$$

Hence $\mathbf{w}_1 = \mathbf{w}_1'$, and consequently $\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = \mathbf{u} - \mathbf{w}_1' = \mathbf{w}_2'$.

■

Theorem 4.1.2: Properties of the Euclidean Inner Product

Statement

> If $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^n$ and k is any scalar ($k \in \mathbb{R}$), then:
>
> - **(a) Symmetry Property:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
> - **(b) Additivity / Distributive Property:** $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
> - **(c) Homogeneity / Scalar Property:** $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$
> - **(d) Positivity Property:** $\mathbf{v} \cdot \mathbf{v} \geq 0$. Further, $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Let the vectors in $\mathbb{R}^n$ be denoted by their components:

$$\mathbf{u} = (u_1, u_2, \ldots, u_n), \quad \mathbf{v} = (v_1, v_2, \ldots, v_n), \quad \mathbf{w} = (w_1, w_2, \ldots, w_n)$$

By definition, the Euclidean inner product (dot product) is:

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Proof of Part (a): $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Symmetry)

By definition of the Euclidean inner product:

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Since multiplication of real numbers is commutative ($u_i v_i = v_i u_i$ for all real numbers u_i, v_i):

$$= v_1 u_1 + v_2 u_2 + \cdots + v_n u_n = \mathbf{v} \cdot \mathbf{u}$$

Proof of Part (b): $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$ (Additivity)

First, compute the vector addition:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1,\ u_2 + v_2,\ \ldots,\ u_n + v_n)$$

Compute the inner product with $\mathbf{w}$:

$$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (u_1 + v_1)w_1 + (u_2 + v_2)w_2 + \cdots + (u_n + v_n)w_n$$

By the distributive property of real numbers ($(u_i + v_i)w_i = u_i w_i + v_i w_i$):

$$= (u_1 w_1 + v_1 w_1) + (u_2 w_2 + v_2 w_2) + \cdots + (u_n w_n + v_n w_n)$$

Regrouping terms:

$$= (u_1 w_1 + u_2 w_2 + \cdots + u_n w_n) + (v_1 w_1 + v_2 w_2 + \cdots + v_n w_n) = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$$

Proof of Part (c): $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$ (Homogeneity)

Compute scalar multiplication: $k\mathbf{u} = (ku_1,\ ku_2,\ \ldots,\ ku_n)$.

$$(k\mathbf{u}) \cdot \mathbf{v} = (ku_1)v_1 + (ku_2)v_2 + \cdots + (ku_n)v_n$$

By associativity of real multiplication ($(ku_i)v_i = k(u_i v_i)$):

$$= k(u_1 v_1) + k(u_2 v_2) + \cdots + k(u_n v_n) = k(u_1 v_1 + u_2 v_2 + \cdots + u_n v_n) = k(\mathbf{u} \cdot \mathbf{v})$$

Proof of Part (d): $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0 \iff \mathbf{v} = \mathbf{0}$ (Positivity)

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2$$

1. **Non-negativity:** Since $v_i^2 \geq 0$ for every real number v_i, the sum of squares is ≥ 0.
2. **Zero condition:** $v_1^2 + v_2^2 + \cdots + v_n^2 = 0 \iff v_1 = v_2 = \cdots = v_n = 0 \iff \mathbf{v} = \mathbf{0}$.

■

Theorem 4.1.3: The Cauchy-Schwarz Inequality in $\mathbb{R}^n$

Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then:
>
> $$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$$
>
> Moreover, equality holds ($|\mathbf{u} \cdot \mathbf{v}| = \|\mathbf{u}\|\|\mathbf{v}\|$) if and only if one vector is a scalar multiple of the other (i.e., the vectors are collinear / linearly dependent).

Proof (via Non-Negative Quadratic Polynomial)

Case 1: $\mathbf{u} = \mathbf{0}$ If $\mathbf{u} = \mathbf{0}$, then $\mathbf{u} \cdot \mathbf{v} = \mathbf{0} \cdot \mathbf{v} = 0$ and $\|\mathbf{u}\| = 0$. Both sides equal 0: $0 \leq 0$, so the inequality (and equality) holds trivially.

Case 2: $\mathbf{u} \neq \mathbf{0}$ Let $t \in \mathbb{R}$ be any real scalar, and consider the vector $t\mathbf{u} + \mathbf{v}$. By the positivity property of the Euclidean inner product (Theorem 4.1.2(d)):

$$p(t) = \|t\mathbf{u} + \mathbf{v}\|^2 = (t\mathbf{u} + \mathbf{v}) \cdot (t\mathbf{u} + \mathbf{v}) \geq 0 \quad \text{for all } t \in \mathbb{R}$$

Expanding using the additivity, homogeneity, and symmetry properties (Theorem 4.1.2):

$$p(t) = (\mathbf{u} \cdot \mathbf{u})t^2 + 2(\mathbf{u} \cdot \mathbf{v})t + (\mathbf{v} \cdot \mathbf{v}) \geq 0$$

Define constants $a = \mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$, $b = 2(\mathbf{u} \cdot \mathbf{v})$, and $c = \mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$. Since $\mathbf{u} \neq \mathbf{0}$, $a = \|\mathbf{u}\|^2 > 0$. The function $p(t) = at^2 + bt + c$ is an upward-opening quadratic polynomial that is **never negative** ($p(t) \geq 0$ for all $t \in \mathbb{R}$).

Therefore, $p(t) = 0$ has **at most one real root** (either zero roots or one repeated root), which means its discriminant $\Delta = b^2 - 4ac$ must satisfy $\Delta \leq 0$:

$$b^2 - 4ac \leq 0$$

$$\big(2(\mathbf{u} \cdot \mathbf{v})\big)^2 - 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \leq 0$$

$$4(\mathbf{u} \cdot \mathbf{v})^2 \leq 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2$$

Dividing by 4:

$$(\mathbf{u} \cdot \mathbf{v})^2 \leq \|\mathbf{u}\|^2\|\mathbf{v}\|^2$$

Taking the non-negative square root of both sides:

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$$

Equality Condition:

When $\mathbf{u} \neq \mathbf{0}$, equality holds if and only if $\Delta = 0$, which means there exists a real root t_0 such that $p(t_0) = \|t_0\mathbf{u} + \mathbf{v}\|^2 = 0$. By positivity, $\|t_0\mathbf{u} + \mathbf{v}\| = 0 \iff t_0\mathbf{u} + \mathbf{v} = \mathbf{0} \iff \mathbf{v} = -t_0\mathbf{u}$ (a scalar multiple of $\mathbf{u}$).

■

Classic Exam Application: Matrix Cauchy-Schwarz Inequality

> **Problem:** If $\mathbf{u}, \mathbf{v}$ are $n \times 1$ column vectors and A is an $n \times n$ matrix, prove:
>
> $$(\mathbf{v}^T A^T A\mathbf{u})^2 \leq (\mathbf{u}^T A^T A\mathbf{u})(\mathbf{v}^T A^T A\mathbf{v})$$

Proof: Let $\mathbf{x} = A\mathbf{u}$ and $\mathbf{y} = A\mathbf{v}$. Notice that $\mathbf{x} \cdot \mathbf{y} = \mathbf{y}^T\mathbf{x} = (A\mathbf{v})^T(A\mathbf{u}) = \mathbf{v}^T A^T A\mathbf{u}$. By the Cauchy-Schwarz Inequality in $\mathbb{R}^n$ ($|\mathbf{x} \cdot \mathbf{y}|^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$):

$$(\mathbf{x} \cdot \mathbf{y})^2 \leq (\mathbf{x} \cdot \mathbf{x})(\mathbf{y} \cdot \mathbf{y})$$

Substituting $\mathbf{x} = A\mathbf{u}$ and $\mathbf{y} = A\mathbf{v}$ back in:

$$\mathbf{x} \cdot \mathbf{x} = (A\mathbf{u})^T(A\mathbf{u}) = \mathbf{u}^T A^T A\mathbf{u}, \qquad \mathbf{y} \cdot \mathbf{y} = (A\mathbf{v})^T(A\mathbf{v}) = \mathbf{v}^T A^T A\mathbf{v}$$

$$(\mathbf{v}^T A^T A\mathbf{u})^2 \leq (\mathbf{u}^T A^T A\mathbf{u})(\mathbf{v}^T A^T A\mathbf{v}) \quad ■$$

Theorem 4.1.4: Properties of Length (Norm) in $\mathbb{R}^n$

Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$ and k is any scalar ($k \in \mathbb{R}$), then:
>
> - **(a) Non-negativity:** $\|\mathbf{u}\| \geq 0$
> - **(b) Zero Condition:** $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$
> - **(c) Homogeneity:** $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$
> - **(d) Triangle Inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$. By definition, $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$.

Proof of Part (a): $\|\mathbf{u}\| \geq 0$

Since $u_i^2 \geq 0$ for each i, $u_1^2 + \cdots + u_n^2 \geq 0$. The principal square root of a non-negative real number is always non-negative, so $\|\mathbf{u}\| \geq 0$.

Proof of Part (b): $\|\mathbf{u}\| = 0 \iff \mathbf{u} = \mathbf{0}$

$$\|\mathbf{u}\| = 0 \iff \sqrt{u_1^2 + \cdots + u_n^2} = 0 \iff u_1^2 + \cdots + u_n^2 = 0 \iff u_1 = \cdots = u_n = 0 \iff \mathbf{u} = \mathbf{0}$$

Proof of Part (c): $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$

$$\|k\mathbf{u}\| = \sqrt{(ku_1)^2 + \cdots + (ku_n)^2} = \sqrt{k^2(u_1^2 + \cdots + u_n^2)} = \sqrt{k^2}\sqrt{u_1^2 + \cdots + u_n^2} = |k|\|\mathbf{u}\|$$

Proof of Part (d): $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (The Triangle Inequality)

Start with the left-hand side squared using $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$:

$$\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2$$

Since any real number is $\leq$ its absolute value ($x \leq |x|$):

$$\leq \|\mathbf{u}\|^2 + 2|\mathbf{u} \cdot \mathbf{v}| + \|\mathbf{v}\|^2$$

Apply the **Cauchy-Schwarz Inequality** (Theorem 4.1.3, $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$):

$$\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$$

Thus $\|\mathbf{u} + \mathbf{v}\|^2 \leq (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$. Since both sides are non-negative, taking square roots yields:

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

■

Theorem 4.1.5: Properties of Distance in $\mathbb{R}^n$

Statement

> If $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^n$, and $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$, then:
>
> - **(a) Non-negativity:** $d(\mathbf{u}, \mathbf{v}) \geq 0$
> - **(b) Zero Condition / Identity:** $d(\mathbf{u}, \mathbf{v}) = 0 \iff \mathbf{u} = \mathbf{v}$
> - **(c) Symmetry:** $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$
> - **(d) Triangle Inequality:** $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$

Proof of Part (a): $d(\mathbf{u}, \mathbf{v}) \geq 0$

Let $\mathbf{x} = \mathbf{u} - \mathbf{v}$. By Theorem 4.1.4(a), $\|\mathbf{x}\| \geq 0 \implies d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| \geq 0$.

Proof of Part (b): $d(\mathbf{u}, \mathbf{v}) = 0 \iff \mathbf{u} = \mathbf{v}$

By Theorem 4.1.4(b), $d(\mathbf{u}, \mathbf{v}) = 0 \iff \|\mathbf{u} - \mathbf{v}\| = 0 \iff \mathbf{u} - \mathbf{v} = \mathbf{0} \iff \mathbf{u} = \mathbf{v}$.

Proof of Part (c): $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(-1)(\mathbf{v} - \mathbf{u})\| \overset{\text{Thm 4.1.4(c)}}{=} |-1|\|\mathbf{v} - \mathbf{u}\| = \|\mathbf{v} - \mathbf{u}\| = d(\mathbf{v}, \mathbf{u})$$

Proof of Part (d): $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$

Add and subtract $\mathbf{w}$ inside the norm:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\|$$

By the Triangle Inequality for norms (Theorem 4.1.4(d)):

$$\|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\| \leq \|\mathbf{u} - \mathbf{w}\| + \|\mathbf{w} - \mathbf{v}\| = d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$$

■

Theorem 4.1.6: The Polarization Identity in $\mathbb{R}^n$

Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$ with the Euclidean inner product, then:
>
> $$\mathbf{u} \cdot \mathbf{v} = \frac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2$$

Proof

Expand both squared norms using $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$ and Theorem 4.1.2:

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v}) \cdot (\mathbf{u}+\mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \quad \text{— (1)}$$

$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v}) \cdot (\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \quad \text{— (2)}$$

Subtract Equation (2) from Equation (1):

$$\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4(\mathbf{u} \cdot \mathbf{v})$$

Dividing by 4 yields:

$$\mathbf{u} \cdot \mathbf{v} = \frac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2$$

■

- **Related Identity — The Parallelogram Equation:** Adding (1) and (2) yields:

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

Theorem 4.1.7: Pythagorean Theorem in $\mathbb{R}^n$

Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors in $\mathbb{R}^n$ ($\mathbf{u} \cdot \mathbf{v} = 0$), then:
>
> $$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Proof

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v}) \cdot (\mathbf{u}+\mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2$$

Since $\mathbf{u} \perp \mathbf{v}$, by definition $\mathbf{u} \cdot \mathbf{v} = 0$:

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2(0) + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

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Part II: General Vector Spaces & Subspaces

Theorem 5.1.1: Fundamental Vector Space Arithmetic

Statement

> Let V be any vector space, $\mathbf{u} \in V$, and $k \in \mathbb{R}$. Using only the 10 vector space axioms, prove:
>
> - **(a)** $0\mathbf{u} = \mathbf{0}$
> - **(b)** $k\mathbf{0} = \mathbf{0}$
> - **(c)** $(-1)\mathbf{u} = -\mathbf{u}$
> - **(d)** If $k\mathbf{u} = \mathbf{0}$, then $k = 0$ or $\mathbf{u} = \mathbf{0}$.

Proofs

- **(a) Proof of $0\mathbf{u} = \mathbf{0}$:** Using the scalar distributive axiom (Axiom 8) with $0 + 0 = 0$:

$$0\mathbf{u} = (0+0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$$

Add the additive inverse $-(0\mathbf{u})$ (Axiom 5) to both sides:

$$0\mathbf{u} + \big(-(0\mathbf{u})\big) = (0\mathbf{u} + 0\mathbf{u}) + \big(-(0\mathbf{u})\big)$$

Using associativity (Axiom 3) and additive identity (Axiom 4):

$$\mathbf{0} = 0\mathbf{u} + \mathbf{0} \implies \mathbf{0} = 0\mathbf{u}$$

- **(b) Proof of $k\mathbf{0} = \mathbf{0}$:** Using $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and Axiom 7:

$$k\mathbf{0} = k(\mathbf{0} + \mathbf{0}) = k\mathbf{0} + k\mathbf{0}$$

Adding $-(k\mathbf{0})$ to both sides yields $\mathbf{0} = k\mathbf{0}$.

- **(c) Proof of $(-1)\mathbf{u} = -\mathbf{u}$:** We show that $\mathbf{u} + (-1)\mathbf{u} = \mathbf{0}$:

$$\mathbf{u} + (-1)\mathbf{u} \overset{\text{Ax 10}}{=} 1\mathbf{u} + (-1)\mathbf{u} \overset{\text{Ax 8}}{=} (1 + (-1))\mathbf{u} = 0\mathbf{u} \overset{\text{Part (a)}}{=} \mathbf{0}$$

Since adding $(-1)\mathbf{u}$ to $\mathbf{u}$ gives $\mathbf{0}$, and the additive inverse of $\mathbf{u}$ is unique, $(-1)\mathbf{u} = -\mathbf{u}$.

- **(d) Proof of Zero-Product Property:** Assume $k\mathbf{u} = \mathbf{0}$. If $k = 0$, the statement holds. Suppose $k \neq 0$; then $\frac{1}{k} \in \mathbb{R}$ exists. Multiply both sides by $\frac{1}{k}$:

$$\frac{1}{k}(k\mathbf{u}) = \frac{1}{k}\mathbf{0}$$

$$\left(\frac{1}{k} \cdot k\right)\mathbf{u} = \mathbf{0} \quad (\text{by Axiom 9 and Part (b)})$$

$$1\mathbf{u} = \mathbf{0} \implies \mathbf{u} = \mathbf{0} \quad (\text{by Axiom 10})$$

■

Theorem 5.2.1: Subspace Test & Intersection vs. Union of Subspaces

Statement (The Subspace Test)

> A non-empty subset W of a vector space V is a **subspace** of V if and only if:
>
> 1. **Contains Zero (Non-empty anchor):** $\mathbf{0} \in W$
> 2. **Closed under Vector Addition:** If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$
> 3. **Closed under Scalar Multiplication:** If $k \in \mathbb{R}$ and $\mathbf{u} \in W$, then $k\mathbf{u} \in W$

Proof: The Intersection of Two Subspaces is Always a Subspace

> **Proposition:** If W_1 and W_2 are subspaces of a vector space V, then their intersection $W_1 \cap W_2$ is also a subspace of V. *(However, their union $W_1 \cup W_2$ is NOT a subspace in general).*

1. **Zero Vector ($\mathbf{0} \in W_1 \cap W_2$):** Since W_1 and W_2 are both subspaces of V, $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2$. By definition of set intersection, $\mathbf{0} \in W_1 \cap W_2$.
2. **Closed under Addition:** Let $\mathbf{u}, \mathbf{v} \in W_1 \cap W_2$. Then:
   - $\mathbf{u}, \mathbf{v} \in W_1 \implies \mathbf{u} + \mathbf{v} \in W_1$ (since W_1 is a subspace).
   - $\mathbf{u}, \mathbf{v} \in W_2 \implies \mathbf{u} + \mathbf{v} \in W_2$ (since W_2 is a subspace). Since $\mathbf{u} + \mathbf{v}$ belongs to both W_1 and W_2, $\mathbf{u} + \mathbf{v} \in W_1 \cap W_2$.
3. **Closed under Scalar Multiplication:** Let $\mathbf{u} \in W_1 \cap W_2$ and $k \in \mathbb{R}$.
   - $\mathbf{u} \in W_1 \implies k\mathbf{u} \in W_1$, and $\mathbf{u} \in W_2 \implies k\mathbf{u} \in W_2$. Hence $k\mathbf{u} \in W_1 \cap W_2$.

Therefore, $W_1 \cap W_2$ is a valid subspace of V. ■

- **Why Union Fails (Counterexample):** In $\mathbb{R}^2$, let $W_1 = \text{the } x\text{-axis}$ and $W_2 = \text{the } y\text{-axis}$. Both are subspaces. Then $(1, 0) \in W_1 \cup W_2$ and $(0, 1) \in W_1 \cup W_2$, yet their sum $(1, 0) + (0, 1) = (1, 1)$ lies on neither axis, so $(1, 1) \notin W_1 \cup W_2$. *(Wait: $W_1 \cup W_2$ is a subspace if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$).*

Theorem 5.2.2: Solution Space of a Homogeneous System is a Subspace

Statement

> If $A\mathbf{x} = \mathbf{0}$ is a homogeneous linear system of m equations in n unknowns, then the set of solution vectors $W = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$ (the Null Space of A) is a subspace of $\mathbb{R}^n$.

Proof

Let $W = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$. We verify the three subspace conditions:

1. **Non-empty / Contains the Zero Vector ($\mathbf{0} \in W$):** Since $A\mathbf{0} = \mathbf{0}$, the zero vector $\mathbf{0} \in \mathbb{R}^n$ is always a solution, so $\mathbf{0} \in W$.
2. **Closure under Vector Addition:** Let $\mathbf{x}, \mathbf{x}' \in W$, so $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{x}' = \mathbf{0}$.

   $$A(\mathbf{x} + \mathbf{x}') = A\mathbf{x} + A\mathbf{x}' = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

   Thus $\mathbf{x} + \mathbf{x}' \in W$.
3. **Closure under Scalar Multiplication:** Let $\mathbf{x} \in W$ (so $A\mathbf{x} = \mathbf{0}$) and let $k \in \mathbb{R}$.

   $$A(k\mathbf{x}) = k(A\mathbf{x}) = k(\mathbf{0}) = \mathbf{0}$$

   Thus $k\mathbf{x} \in W$.

Since W contains $\mathbf{0}$ and is closed under addition and scalar multiplication, W **is a subspace of** $\mathbb{R}^n$.

■

- **Non-Homogeneous Trap ($A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$):** NEVER a subspace because $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$ ($\mathbf{0}$ is missing).

Theorem 5.2.3: $\mathrm{Span}(S)$ is the Smallest Subspace Containing S

Statement

> If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then:
>
> - **(a)** The set $W = \mathrm{span}(S)$ is a subspace of V.
> - **(b)** W is the *smallest* subspace of V that contains S; that is, any other subspace W' of V containing S must satisfy $W \subseteq W'$.

Proof

- **Part (a): $W = \mathrm{span}(S)$ is a subspace of V**

  1. **Zero Vector:** Setting $c_1 = c_2 = \cdots = c_k = 0$ gives $\mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_k \in \mathrm{span}(S) = W$.
  2. **Closed under Addition:** Let $\mathbf{u} = \sum_{i=1}^{k} c_i\mathbf{v}_i$ and $\mathbf{w} = \sum_{i=1}^{k} d_i\mathbf{v}_i$ be in W. Then:

     $$\mathbf{u} + \mathbf{w} = (c_1 + d_1)\mathbf{v}_1 + \cdots + (c_k + d_k)\mathbf{v}_k$$

     Since each $c_i + d_i \in \mathbb{R}$, $\mathbf{u} + \mathbf{w}$ is a linear combination of vectors in S, so $\mathbf{u} + \mathbf{w} \in W$.
  3. **Closed under Scalar Multiplication:** Let $\alpha \in \mathbb{R}$ and $\mathbf{u} = \sum_{i=1}^{k} c_i\mathbf{v}_i \in W$. Then:

     $$\alpha\mathbf{u} = (\alpha c_1)\mathbf{v}_1 + \cdots + (\alpha c_k)\mathbf{v}_k \in W$$

  Thus $W = \mathrm{span}(S)$ is a subspace of V.

- **Part (b): W is the smallest subspace containing S** First, each $\mathbf{v}_j \in W$ (set $c_j = 1$ and all other $c_i = 0$), so $S \subseteq W$. Now let W' be *any* subspace of V such that $S \subseteq W'$. That means $\mathbf{v}_1, \ldots, \mathbf{v}_k \in W'$. Because W' is a subspace, it is closed under scalar multiplication and addition. Therefore, for any scalars $c_1, \ldots, c_k \in \mathbb{R}$, the linear combination:

  $$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k \in W'$$

  Since every element of $W = \mathrm{span}(S)$ has this form, $W \subseteq W'$.

■

Theorem: The Column Space $\mathrm{Col}(A)$ is Always a Subspace of $\mathbb{R}^m$

Statement

> For any $m \times n$ matrix A, the column space $\mathrm{Col}(A) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}$ is a valid subspace of $\mathbb{R}^m$.

Proof

1. **Zero Vector ($\mathbf{0}_m \in \text{Col}(A)$):** Choose $\mathbf{x} = \mathbf{0}_n \in \mathbb{R}^n$. Since $A\mathbf{0}_n = \mathbf{0}_m$, $\mathbf{0}_m \in \text{Col}(A)$.
2. **Closed under Addition:** Let $\mathbf{u}, \mathbf{v} \in \text{Col}(A)$. Then there exist $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$ such that $\mathbf{u} = A\mathbf{x}_1$ and $\mathbf{v} = A\mathbf{x}_2$.

   $$\mathbf{u} + \mathbf{v} = A\mathbf{x}_1 + A\mathbf{x}_2 = A(\mathbf{x}_1 + \mathbf{x}_2)$$

   Since $(\mathbf{x}_1 + \mathbf{x}_2) \in \mathbb{R}^n$, $\mathbf{u} + \mathbf{v} \in \text{Col}(A)$.
3. **Closed under Scalar Multiplication:** Let $\mathbf{u} = A\mathbf{x}_1 \in \text{Col}(A)$ and $c \in \mathbb{R}$.

   $$c\mathbf{u} = c(A\mathbf{x}_1) = A(c\mathbf{x}_1)$$

   Since $c\mathbf{x}_1 \in \mathbb{R}^n$, $c\mathbf{u} \in \text{Col}(A)$.

■

Theorem: Consistency Theorem for Linear Systems ($A\mathbf{x} = \mathbf{b}$)

Statement

> A system of linear equations $A\mathbf{x} = \mathbf{b}$ is **consistent** if and only if the vector $\mathbf{b}$ is in the column space of A:
>
> $$A\mathbf{x} = \mathbf{b} \text{ is consistent } \iff \mathbf{b} \in \text{Col}(A)$$

Proof (Bidirectional $\iff$)

Let $A = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{bmatrix}$ be $m \times n$ with $\mathbf{c}_j \in \mathbb{R}^m$, and $\mathbf{x} = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T \in \mathbb{R}^n$. By definition of matrix multiplication:

$$A\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n \quad \text{— (Eq. 1)}$$

- **Forward Direction ($\implies$):** Assume $A\mathbf{x} = \mathbf{b}$ is consistent. There exists at least one vector $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$. By Eq. 1, $x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \mathbf{b}$. Thus $\mathbf{b}$ is a linear combination of the columns of A, meaning $\mathbf{b} \in \text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\} = \text{Col}(A)$.
- **Backward Direction ($\impliedby$):** Assume $\mathbf{b} \in \text{Col}(A)$. By definition of column space, there exist scalars $s_1, \dots, s_n \in \mathbb{R}$ such that $s_1\mathbf{c}_1 + \cdots + s_n\mathbf{c}_n = \mathbf{b}$. Defining $\mathbf{s} = \begin{bmatrix} s_1 & \dots & s_n \end{bmatrix}^T \in \mathbb{R}^n$, Eq. 1 gives $A\mathbf{s} = \mathbf{b}$. Thus the system has a solution and is consistent.

■

Part III: Linear Independence, Span, Bases & Dimension

Theorem 5.3.1: The Redundancy Equivalence Theorem

Statement

> A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ with $k \geq 2$ vectors is **linearly dependent** if and only if at least one vector in S is expressible as a linear combination of the other vectors in S.

Proof (Bidirectional $\iff$)

- **Direction 1 ($\impliedby$): Redundancy $\implies$ Linear Dependence** Suppose at least one vector in S (say $\mathbf{v}_1$, without loss of generality) can be written as a linear combination of the others:

  $$\mathbf{v}_1 = a_2\mathbf{v}_2 + a_3\mathbf{v}_3 + \cdots + a_k\mathbf{v}_k$$

  Rearranging all terms to one side:

  $$(-1)\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$$

  Since the coefficient of $\mathbf{v}_1$ is $-1 \neq 0$, this is a non-trivial solution to $\sum c_i\mathbf{v}_i = \mathbf{0}$. Hence S is linearly dependent.
- **Direction 2 ($\implies$): Linear Dependence $\implies$ Redundancy** Suppose S is linearly dependent. Then there exist scalars $c_1, \dots, c_k$, **not all zero**, such that:

  $$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

Since not all c_i are zero, pick an index with a non-zero scalar (without loss of generality, say $c_1 \neq 0$). Isolate $c_1\mathbf{v}_1$ and divide by c_1:

$$c_1\mathbf{v}_1 = -c_2\mathbf{v}_2 - \cdots - c_k\mathbf{v}_k$$

$$\mathbf{v}_1 = \left(-\frac{c_2}{c_1}\right)\mathbf{v}_2 + \cdots + \left(-\frac{c_k}{c_1}\right)\mathbf{v}_k$$

Thus $\mathbf{v}_1$ is expressible as a linear combination of the remaining vectors.

■

Theorem 5.3.2: Zero Vector and Two-Vector Dependency

Statement

> - **(a)** Any finite set of vectors containing the zero vector $\mathbf{0}$ is **linearly dependent**.
> - **(b)** A set $S = \{\mathbf{s}_1, \mathbf{s}_2\}$ with exactly two vectors is **linearly independent** if and only if neither vector is a scalar multiple of the other.

Proofs

- **Proof of Part (a):** Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r, \mathbf{0}\}$. Consider the linear combination:

  $$0\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_r + 1\mathbf{0} = \mathbf{0}$$

  Because the coefficient of $\mathbf{0}$ is $1 \neq 0$, the equation has a **non-trivial solution** (not all coefficients are zero). Therefore, S is linearly dependent.

- **Proof of Part (b) (via Contrapositive):** Proving *"S is linearly independent iff neither vector is a scalar multiple of the other"* is logically equivalent to proving its contrapositive: *"S is linearly dependent iff one vector is a scalar multiple of the other."*

  - **($\Longrightarrow$) Assume S is linearly dependent:** There exist scalars k_1, k_2, not both zero ($k_1 \neq 0$ or $k_2 \neq 0$), such that:

    $$k_1\mathbf{s}_1 + k_2\mathbf{s}_2 = \mathbf{0}$$

    - If $k_1 \neq 0$: $\mathbf{s}_1 = -\frac{k_2}{k_1}\mathbf{s}_2$ ($\mathbf{s}_1$ is a scalar multiple of $\mathbf{s}_2$).
    - If $k_2 \neq 0$: $\mathbf{s}_2 = -\frac{k_1}{k_2}\mathbf{s}_1$ ($\mathbf{s}_2$ is a scalar multiple of $\mathbf{s}_1$).

  - **($\Longleftarrow$) Assume one vector is a scalar multiple of the other:** Without loss of generality, suppose $\mathbf{s}_2 = k\mathbf{s}_1$ for some scalar $k \in \mathbb{R}$. Then:

    $$k\mathbf{s}_1 - 1\mathbf{s}_2 = \mathbf{0}$$

    Since the coefficient of $\mathbf{s}_2$ is $-1 \neq 0$, the coefficients are not all zero, so S is linearly dependent.

■

Theorem: Every Non-Empty Subset of a Linearly Independent Set is Linearly Independent

Statement

> Show that if $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$ is a linearly independent set of vectors in V, then every non-empty subset $S' \subseteq S$ is also linearly independent.

Proof

Let $S' = \{\mathbf{v}_{i_1}, \mathbf{v}_{i_2}, \ldots, \mathbf{v}_{i_k}\}$ be any non-empty subset of S (where $1 \leq k \leq r$). Consider the homogeneous linear combination of vectors in S':

$$c_1\mathbf{v}_{i_1} + c_2\mathbf{v}_{i_2} + \cdots + c_k\mathbf{v}_{i_k} = \mathbf{0}$$

We can expand this equation to include **all** vectors in S by assigning a coefficient of 0 to every vector in $S \setminus S'$:

$$c_1\mathbf{v}_{i_1} + \cdots + c_k\mathbf{v}_{i_k} + \sum_{\mathbf{v}_j \in S \setminus S'} 0 \cdot \mathbf{v}_j = \mathbf{0}$$

This forms a linear combination of the entire set S that equals $\mathbf{0}$. Since S is linearly independent, **every scalar coefficient in this equation must be zero**. In particular:

$$c_1 = c_2 = \cdots = c_k = 0$$

Thus, S' is linearly independent.

■

Theorem: The Plus/Minus Theorem (Extending & Trimming Sets)

Statement

> Let S be a non-empty set of vectors in a vector space V:
>
> - **(a) The "Plus" Part (Extending Independence):** If $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is a linearly independent set and $\mathbf{v} \in V$ is a vector that **does NOT lie in** $\text{span}(S)$, then $S \cup \{\mathbf{v}\} = \{\mathbf{v}_1, \ldots, \mathbf{v}_r, \mathbf{v}\}$ is also **linearly independent**.
> - **(b) The "Minus" Part (Trimming Redundancy):** If a vector $\mathbf{v} \in S$ is expressible as a linear combination of the other vectors in S, and if $S \setminus \{\mathbf{v}\}$ denotes the set obtained by removing $\mathbf{v}$ from S, then S and $S \setminus \{\mathbf{v}\}$ span the **exact same space**:
>
> $$\text{span}(S) = \text{span}(S \setminus \{\mathbf{v}\})$$

Proofs

- **Proof of (a) — Plus Part (illustrated for $\{\mathbf{v}_1, \ldots, \mathbf{v}_r, \mathbf{v}\}$):** Consider the homogeneous equation:

  $$k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r + k\mathbf{v} = \mathbf{0}$$

  Suppose for contradiction that $k \neq 0$. Then we can isolate $\mathbf{v}$ and divide by k:

  $$\mathbf{v} = \left(-\frac{k_1}{k}\right)\mathbf{v}_1 + \cdots + \left(-\frac{k_r}{k}\right)\mathbf{v}_r$$

  This expresses $\mathbf{v}$ as a linear combination of vectors in S, contradicting the given hypothesis that $\mathbf{v} \notin \text{span}(S)$. Therefore, we must have $k = 0$. Substituting $k = 0$ back leaves:

  $$k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r = \mathbf{0}$$

  Since S is linearly independent, $k_1 = k_2 = \cdots = k_r = 0$. Since all coefficients $(k_1, \ldots, k_r, k)$ are zero, $S \cup \{\mathbf{v}\}$ is linearly independent.

- **Proof of (b) — Minus Part:** Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_r, \mathbf{v}\}$ where $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_r\mathbf{v}_r$.

  1. Clearly $\text{span}(S \setminus \{\mathbf{v}\}) \subseteq \text{span}(S)$ since removing a vector cannot add combinations.
  2. Conversely, let $\mathbf{w} \in \text{span}(S)$:

     $$\mathbf{w} = c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r + c\mathbf{v}$$

     Substitute $\mathbf{v} = \sum_{i=1}^{r} a_i\mathbf{v}_i$:

     $$\mathbf{w} = (c_1 + ca_1)\mathbf{v}_1 + \cdots + (c_r + ca_r)\mathbf{v}_r \in \text{span}(S \setminus \{\mathbf{v}\})$$

     Thus $\text{span}(S) \subseteq \text{span}(S \setminus \{\mathbf{v}\})$, proving equality.

■

Theorem 5.3.3 & 5.4.3: Dimension Bounds on Linear Independence and Spanning

Statement (General Vector Space V with $\dim(V) = n$)

> Let V be a finite-dimensional vector space and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be any basis for V (so $\dim(V) = n$). Prove that:
>
> - **(a)** Any set $S' = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\} \subset V$ with $m > n$ **vectors** is **linearly dependent**. *(In $\mathbb{R}^n$, any $> n$ vectors are LD).*
> - **(b)** Any set $S' = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\} \subset V$ with $m < n$ **vectors cannot span** V.

Proofs

Proof of Part (a): More than n vectors ($m > n$) $\implies$ Linearly Dependent

Since $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis of V, every vector $\mathbf{w}_j \in S'$ can be written as a linear combination of the basis vectors:

$$\mathbf{w}_j = \sum_{i=1}^{n} a_{ij}\mathbf{v}_i \quad \text{for } j = 1, 2, \ldots, m$$

To test linear independence of S', consider:

$$k_1\mathbf{w}_1 + k_2\mathbf{w}_2 + \cdots + k_m\mathbf{w}_m = \mathbf{0}$$

Substituting the basis expansions and regrouping by $\mathbf{v}_1, \ldots, \mathbf{v}_n$:

$$\sum_{j=1}^{m} k_j \left(\sum_{i=1}^{n} a_{ij}\mathbf{v}_i \right) = \sum_{i=1}^{n} \left(\sum_{j=1}^{m} a_{ij}k_j \right) \mathbf{v}_i = \mathbf{0}$$

Because $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, every coefficient multiplying $\mathbf{v}_i$ must equal 0:

$$\begin{cases} a_{11}k_1 + a_{12}k_2 + \cdots + a_{1m}k_m = 0 \\ \vdots \\ a_{n1}k_1 + a_{n2}k_2 + \cdots + a_{nm}k_m = 0 \end{cases}$$

This is a homogeneous linear system of n **equations in** m **unknowns** ($k_1, \ldots, k_m$). Since $m > n$ (more unknowns than equations), the system must have at least one free variable and thus admits a **non-trivial solution**. Therefore, S' is linearly dependent!

(Note: For $V = \mathbb{R}^n$, the standard basis is $\mathbf{e}_1, \ldots, \mathbf{e}_n$, so the coefficients a_{ij} are simply the entries of the vectors $\mathbf{w}_j$, giving the exact $n \times m$ matrix system directly!)

Proof of Part (b): Fewer than n vectors ($m < n$) $\implies$ Does Not Span V

Suppose for contradiction that $S' = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ with $m < n$ *does* span V. Then each basis vector $\mathbf{v}_j \in S$ ($j = 1, \ldots, n$) can be written as a linear combination of $\mathbf{w}_1, \ldots, \mathbf{w}_m$:

$$\mathbf{v}_j = \sum_{i=1}^{m} a_{ij}\mathbf{w}_i$$

Consider the homogeneous equation on the basis S: $k_1\mathbf{v}_1 + \cdots + k_n\mathbf{v}_n = \mathbf{0}$. Substituting gives:

$$\sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij}k_j \right) \mathbf{w}_i = \mathbf{0}$$

Setting each inner sum to zero produces a homogeneous system of m **equations in** n **unknowns** ($k_1, \ldots, k_n$):

$$\sum_{j=1}^{n} a_{ij}k_j = 0 \quad (i = 1, \ldots, m)$$

Since $m < n$ (more unknowns than equations), this homogeneous system has a **non-trivial solution** (not all $k_j = 0$) that satisfies $\sum_{j=1}^{n} k_j\mathbf{v}_j = \mathbf{0}$. This directly contradicts the fact that $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis (linearly independent). Therefore, S' cannot span V.

■

Theorem 5.4.1: Uniqueness of Basis Representation

Statement

> If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for a vector space V, then every vector $\mathbf{v} \in V$ can be expressed in the form $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$ in **exactly one way**.

Proof

Since S spans V, at least one such representation exists for every $\mathbf{v} \in V$. To prove uniqueness, suppose $\mathbf{v}$ has two representations:

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n \quad \text{— (1)}$$

$$\mathbf{v} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n \quad \text{— (2)}$$

Subtract Equation (1) from Equation (2):

$$\mathbf{v} - \mathbf{v} = (k_1 - c_1)\mathbf{v}_1 + (k_2 - c_2)\mathbf{v}_2 + \cdots + (k_n - c_n)\mathbf{v}_n$$

$$\mathbf{0} = (k_1 - c_1)\mathbf{v}_1 + (k_2 - c_2)\mathbf{v}_2 + \cdots + (k_n - c_n)\mathbf{v}_n$$

Since S is a basis, its vectors are **linearly independent**, so the only solution is the trivial one where every coefficient is zero:

$$k_1 - c_1 = 0, \quad k_2 - c_2 = 0, \quad \ldots, \quad k_n - c_n = 0$$

$$k_1 = c_1, \quad k_2 = c_2, \quad \ldots, \quad k_n = c_n$$

Thus, the coordinate representation is strictly unique.

■

Theorem 5.4.5: The Half-Test for Basis (Basis Theorem in n-Dimensional Space)

Statement

> If V is an n-dimensional vector space ($\dim(V) = n$) and S is a set in V containing **exactly n vectors**, then S is a basis for V if **EITHER** S spans V **OR** S is linearly independent.

Proof

- **Case 1: Assume $|S| = n$ and S spans V.** *(We must prove S is linearly independent).* Suppose for contradiction that S is NOT linearly independent. By Theorem 5.3.1, some vector $\mathbf{v} \in S$ is a linear combination of the other vectors in S. By the **Minus part** of the Plus/Minus Theorem, removing $\mathbf{v}$ preserves the span, so the remaining set $S \setminus \{\mathbf{v}\}$ (which has $n - 1$ **vectors**) still spans V. However, this is a **contradiction** by Theorem 5.4.3(b): fewer than n vectors ($n - 1 < n$) cannot span an n-dimensional vector space! Hence, S must be linearly independent, and therefore S is a basis for V.
- **Case 2: Assume $|S| = n$ and S is linearly independent.** *(We must prove S spans V).* Suppose for contradiction that S does NOT span V. Then there must exist some vector $\mathbf{v} \in V$ such that $\mathbf{v} \notin \text{span}(S)$. By the **Plus part** of the Plus/Minus Theorem, inserting $\mathbf{v}$ into S yields $S \cup \{\mathbf{v}\}$ (which has $n + 1$ **vectors**) that is also linearly independent. However, this is a **contradiction** by Theorem 5.4.3(a): a set with more than n vectors ($n + 1 > n$) in an n-dimensional space cannot be linearly independent! Hence, S must span V, and therefore S is a basis for V.

■

Part IV: Row Space, Column Space, Rank & Nullity

Theorem: Null Space & Column Linear Independence Equivalence

Statement

> For any $m \times n$ matrix A, the column vectors of A are **linearly independent** if and only if the null space of A contains only the zero vector:
>
> $$\text{Columns of } A \text{ are linearly independent} \iff \text{Null}(A) = \{\mathbf{0}\} \iff \text{Nullity}(A) = 0$$

Proof

Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_n \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \ldots & x_n \end{bmatrix}^T$. By definition of matrix multiplication:

$$A\mathbf{x} = \mathbf{0} \iff x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$$

- **($\Longrightarrow$)** If the columns are linearly independent, the only scalars solving $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ are $x_1 = \cdots = x_n = 0$, i.e. $\mathbf{x} = \mathbf{0}$. Hence $\text{Null}(A) = \{\mathbf{0}\}$.
- **($\Longleftarrow$)** Conversely, if $\text{Null}(A) = \{\mathbf{0}\}$, then the only solution to $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$, meaning $x_1 = \cdots = x_n = 0$ is the only solution to $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$. Thus the columns are linearly independent.

■

Theorems 5.5.4 & 5.5.5: Row Operations Preserve Row Space & Column Relations

Statements

> Let R be obtained from a matrix A by elementary row operations (e.g., R is an REF/RREF of A):
>
> - **(a) Theorem 5.5.4:** Elementary row operations do **not** change the row space of a matrix: $\text{Row}(A) = \text{Row}(R)$ (and thus $\dim(\text{Row}(A)) = \dim(\text{Row}(R))$).
> - **(b) Theorem 5.5.5:** Elementary row operations do **not** change the linear dependency relationships among the columns of A. Therefore, the pivot columns of A form a basis for $\text{Col}(A)$, and $\dim(\text{Col}(A)) = \dim(\text{Col}(R))$.

Proof Sketches

- **Why $\text{Row}(A) = \text{Row}(R)$:** Every elementary row operation (swapping two rows, scaling a row by $k \neq 0$, or replacing R_i by $R_i + cR_j$) produces new rows that are **linear combinations of the rows of A**. Thus $\text{Row}(R) \subseteq \text{Row}(A)$. Because every ERO is reversible by an inverse ERO, the rows of A are also linear combinations of the rows of R, so $\text{Row}(A) \subseteq \text{Row}(R)$. Hence $\text{Row}(A) = \text{Row}(R)$.
- **Why column relations are preserved:** If $R = EA$ for an invertible product of elementary matrices E, then $R\mathbf{x} = \mathbf{0} \iff E(A\mathbf{x}) = \mathbf{0} \iff A\mathbf{x} = \mathbf{0}$. Writing $A\mathbf{x} = \sum x_j \mathbf{a}_j$ and $R\mathbf{x} = \sum x_j \mathbf{r}_j$, the **exact same scalars** x_j satisfy $\sum x_j \mathbf{a}_j = \mathbf{0}$ as satisfy $\sum x_j \mathbf{r}_j = \mathbf{0}$! Therefore, a subset of columns of A is linearly independent (or forms a basis of $\text{Col}(A)$) if and only if the corresponding columns of R are linearly independent!

Theorem 5.6.1: Row Rank Equals Column Rank

Statement

> If A is any $m \times n$ matrix, then the row space and column space of A have the **exact same dimension**:
>
> $$\dim(\text{Row}(A)) = \dim(\text{Col}(A))$$

Proof

Let R be any row-echelon form (REF) of A.

- By Theorem 5.5.4, elementary row operations preserve the row space, so:

$$\dim(\text{Row}(A)) = \dim(\text{Row}(R))$$

- By Theorem 5.5.5, elementary row operations preserve linear independence relations among the columns, so:

$$\dim(\text{Col}(A)) = \dim(\text{Col}(R))$$

Thus, the proof is complete if we show $\dim(\text{Row}(R)) = \dim(\text{Col}(R))$ for the row-echelon form R:

1. The **nonzero row vectors** of R are linearly independent (because of the staircase of leading 1's) and span $\text{Row}(R)$, so:

$$\dim(\text{Row}(R)) = \text{number of nonzero rows in } R$$

2. The **columns containing leading 1's** (pivot columns) of R form a basis for $\text{Col}(R)$, so:

$$\dim(\text{Col}(R)) = \text{number of columns containing leading } 1\text{'s in } R$$

3. However, every nonzero row of R has **exactly one** leading 1, and every column with a leading 1 corresponds to **exactly one** nonzero row. Therefore, the number of nonzero rows equals the number of leading 1's (pivots).

Consequently, $\dim(\text{Row}(R)) = \dim(\text{Col}(R))$, and hence:

$$\dim(\text{Row}(A)) = \dim(\text{Col}(A))$$

■

Theorem 5.6.3: The Rank-Nullity Theorem for Matrices

Statement

> If A is any $m \times n$ matrix (having n columns), then:
>
> $$\text{Rank}(A) + \text{Nullity}(A) = n$$
>
> where $\text{Rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A))$ and $\text{Nullity}(A) = \dim(\text{Null}(A))$.

Proof

Let $R = \text{RREF}(A)$ be the reduced row-echelon form of A, and let r be the number of leading 1's (pivots) in R.

1. By definition of rank and Theorem 5.6.1, $\text{Rank}(A) = r$ (the number of pivot columns).
2. In the homogeneous system $A\mathbf{x} = \mathbf{0} \iff R\mathbf{x} = \mathbf{0}$:
   - There are n total variables (one for each column of A).
   - Exactly r variables correspond to pivot columns (**leading/pivot variables**).
   - The remaining $n - r$ variables correspond to non-pivot columns (**free variables**).
3. Solving $R\mathbf{x} = \mathbf{0}$ expresses each of the r pivot variables uniquely in terms of the $n - r$ free variables. Setting one free variable to 1 and all other free variables to 0 yields $n - r$ solution vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{n-r} \in \mathbb{R}^n$ such that every solution is $\mathbf{x} = t_1\mathbf{v}_1 + \cdots + t_{n-r}\mathbf{v}_{n-r}$.
   - Because each $\mathbf{v}_j$ has a 1 in the coordinate of its free variable and 0 in the coordinates of all other free variables (forming an embedded $(n-r) \times (n-r)$ identity submatrix), these $n - r$ vectors are strictly linearly independent and span $\text{Null}(A)$.
   - *(If $r = n$, there are 0 free variables, $\text{Null}(A) = \{\mathbf{0}\}$, and $\text{Nullity}(A) = 0 = n - n$).*
4. Therefore, $\text{Nullity}(A) = n - r$. Summing the two gives:

$$\text{Rank}(A) + \text{Nullity}(A) = r + (n - r) = n$$

■

Part V: Linear Transformations

Theorem: Fundamental Properties of Linear Transformations

Statement

> Let $T : V \to W$ be a linear transformation ($T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$). Prove:
>
> - **(a) Zero Vector Preservation:** $T(\mathbf{0}_V) = \mathbf{0}_W$
> - **(b) Negation Preservation:** $T(-\mathbf{u}) = -T(\mathbf{u})$
> - **(c) Difference Preservation:** $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$
> - **(d) Superposition Principle:** $T(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$

Proofs

- **(a)** Choose $c = 0 \in \mathbb{R}$ and any $\mathbf{u} \in V$:

$$T(\mathbf{0}_V) = T(0 \cdot \mathbf{u}) \overset{\text{homogeneity}}{=} 0 \cdot T(\mathbf{u}) = \mathbf{0}_W$$

- **(b)** Choose $c = -1$:

$$T(-\mathbf{u}) = T((-1)\mathbf{u}) = (-1)T(\mathbf{u}) = -T(\mathbf{u})$$

- **(c)** Combine additivity and (b):

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u} + (-\mathbf{v})) = T(\mathbf{u}) + T(-\mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$$

- **(d)** Apply additivity repeatedly, then pull out each scalar c_i using homogeneity:

$$T(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = T(c_1\mathbf{v}_1) + \cdots + T(c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$$

■

Theorem: Every Matrix Transformation $T(\mathbf{x}) = A\mathbf{x}$ is Linear

Statement

> For any $m \times n$ matrix A, the transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{x}) = A\mathbf{x}$ is a linear transformation.

Proof

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. By the distributive and associative properties of matrix multiplication:

1. **Additivity:**

$$T(\mathbf{u}+\mathbf{v}) = A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v} = T(\mathbf{u}) + T(\mathbf{v})$$

2. **Homogeneity:**

$$T(c\mathbf{u}) = A(c\mathbf{u}) = c(A\mathbf{u}) = cT(\mathbf{u})$$

Both linearity axioms hold, so T is linear.

■

Theorem: Standard Matrix Representation & Uniqueness

Statement

> Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be any linear transformation. Then there exists a **unique** $m \times n$ matrix $[T]$ (called the standard matrix of T) such that $T(\mathbf{x}) = [T]\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, given explicitly by:
>
> $$[T] = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$
>
> where $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$.

Proof

1. **Existence:** Any vector $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$ can be written as $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$. By linearity of T:

$$T(\mathbf{x}) = T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n)$$

   By definition of matrix-vector multiplication, this linear combination of columns equals:

$$T(\mathbf{x}) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [T]\mathbf{x}$$

2. **Uniqueness:** Suppose another $m \times n$ matrix $B = \begin{bmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_n \end{bmatrix}$ also satisfies $T(\mathbf{x}) = B\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Evaluating at each standard basis vector $\mathbf{e}_j$ ($j = 1, \dots, n$):

$$T(\mathbf{e}_j) = B\mathbf{e}_j = \mathbf{b}_j \quad (\text{since } B\mathbf{e}_j \text{ extracts the } j\text{-th column of } B)$$

   Hence every column of B equals $T(\mathbf{e}_j)$, so $B = [T]$.

■

Theorem: Range of a Matrix Transformation Equals the Column Space

Statement

> Prove that the range of a matrix transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ given by $T(\mathbf{x}) = A\mathbf{x}$ is equal to the column space of A:
>
> $$R(T) = \text{Col}(A)$$

Proof

By definition, the range of T is the set of all possible outputs:

$$R(T) = \{T(\mathbf{x}) \mid \mathbf{x} \in \mathbb{R}^n\}$$

Let A be an $m \times n$ matrix partitioned into its column vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^m$:

$$A = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{bmatrix}, \quad \text{and let } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

Evaluating $T(\mathbf{x}) = A\mathbf{x}$ via the column view of matrix multiplication:

$$T(\mathbf{x}) = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n$$

Thus, every output of T is a linear combination of the columns of A. As $\mathbf{x}$ ranges over every possible input vector $(x_1, \dots, x_n) \in \mathbb{R}^n$, the output $T(\mathbf{x}) = A\mathbf{x}$ ranges over every possible linear combination of the columns of A, which is precisely $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \text{Col}(A)$.

Hence, $R(T) = \text{Col}(A)$.

■

Theorem: Kernel and Range are Subspaces

Statement

> Let $T : V \to W$ be a linear transformation between vector spaces V and W. Prove that:
>
> - **(a)** $\ker(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W\}$ is a **subspace of the domain** V.
> - **(b)** $R(T) = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}$ is a **subspace of the codomain** W.

Proofs

Part (a): $\ker(T)$ is a subspace of V

1. **Contains Zero ($\mathbf{0}_V \in \ker(T)$):** Since T is linear, $T(\mathbf{0}_V) = \mathbf{0}_W$, so $\mathbf{0}_V \in \ker(T)$.
2. **Closed under Addition:** Let $\mathbf{u}, \mathbf{v} \in \ker(T)$, meaning $T(\mathbf{u}) = \mathbf{0}_W$ and $T(\mathbf{v}) = \mathbf{0}_W$.

   $$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$$

   Therefore, $\mathbf{u} + \mathbf{v} \in \ker(T)$.
3. **Closed under Scalar Multiplication:** Let $\mathbf{u} \in \ker(T)$ (so $T(\mathbf{u}) = \mathbf{0}_W$) and $c \in \mathbb{R}$.

   $$T(c\mathbf{u}) = cT(\mathbf{u}) = c\mathbf{0}_W = \mathbf{0}_W$$

   Therefore, $c\mathbf{u} \in \ker(T)$.

Part (b): $R(T)$ is a subspace of W

1. **Contains Zero ($\mathbf{0}_W \in R(T)$):** Since $\mathbf{0}_V \in V$ and $T(\mathbf{0}_V) = \mathbf{0}_W$, there is an input mapping to $\mathbf{0}_W$, so $\mathbf{0}_W \in R(T)$.
2. **Closed under Addition:** Let $\mathbf{w}_1, \mathbf{w}_2 \in R(T)$. Then there exist $\mathbf{v}_1, \mathbf{v}_2 \in V$ such that $T(\mathbf{v}_1) = \mathbf{w}_1$ and $T(\mathbf{v}_2) = \mathbf{w}_2$. By additivity of T:

   $$\mathbf{w}_1 + \mathbf{w}_2 = T(\mathbf{v}_1) + T(\mathbf{v}_2) = T(\mathbf{v}_1 + \mathbf{v}_2)$$

   Since $(\mathbf{v}_1 + \mathbf{v}_2) \in V$, $\mathbf{w}_1 + \mathbf{w}_2$ is the image of an element of V, so $\mathbf{w}_1 + \mathbf{w}_2 \in R(T)$.
3. **Closed under Scalar Multiplication:** Let $\mathbf{w}_1 = T(\mathbf{v}_1) \in R(T)$ and $c \in \mathbb{R}$.

   $$c\mathbf{w}_1 = cT(\mathbf{v}_1) = T(c\mathbf{v}_1)$$

   Since $c\mathbf{v}_1 \in V$, $c\mathbf{w}_1 \in R(T)$.

■

Theorem: The Trivial Kernel Test for One-to-One Linear Transformations

Statement

> Let $T : V \to W$ be a linear transformation between vector spaces V and W. Then T is **one-to-one** (injective) if and only if the kernel of T consists of only the zero vector:
>
> $$T \text{ is one-to-one} \iff \ker(T) = \{\mathbf{0}\}$$

Core Intuition: Why Do We Only Solve $A\mathbf{x} = \mathbf{0}$ Instead of Checking Every Arbitrary $A\mathbf{x} = \mathbf{b}$?

A very natural question is: *"One-to-one means NO output $\mathbf{b}$ anywhere can have two different inputs. Why do we only test $\mathbf{b} = \mathbf{0}$ instead of testing an arbitrary $\mathbf{b}$?"*

The answer is **linearity** ($T(\mathbf{u} - \mathbf{w}) = T(\mathbf{u}) - T(\mathbf{w})$), which works in both directions:

1. **Any collision at ANY $\mathbf{b}$ automatically shows up at $\mathbf{0}$ (The $\mathbf{b}$ Cancels Out!):** Suppose some arbitrary vector $\mathbf{b}$ could be hit by two distinct inputs $\mathbf{u} \neq \mathbf{w}$:

   $$A\mathbf{u} = \mathbf{b} \quad \text{and} \quad A\mathbf{w} = \mathbf{b}$$

   Subtract the two equations:

   $$A\mathbf{u} - A\mathbf{w} = \mathbf{b} - \mathbf{b} \implies A(\mathbf{u} - \mathbf{w}) = \mathbf{0}$$

   The arbitrary $\mathbf{b}$ **completely cancels out**! Every collision $(\mathbf{u}, \mathbf{w})$ at any target $\mathbf{b}$ creates a non-zero vector $\mathbf{v}_h = (\mathbf{u} - \mathbf{w}) \neq \mathbf{0}$ that solves $A\mathbf{x} = \mathbf{0}$. Therefore, if $A\mathbf{x} = \mathbf{0}$ has ONLY $\mathbf{x} = \mathbf{0}$, no collision can exist at *any* $\mathbf{b}$!

2. **Any collapse at $\mathbf{0}$ copies itself to EVERY reachable $\mathbf{b}$ ($\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$):** Conversely, if there exists a non-zero $\mathbf{v}_h \in \ker(T)$ ($A\mathbf{v}_h = \mathbf{0}$), take ANY target $\mathbf{b}$ reached by some input $\mathbf{x}_p$ ($A\mathbf{x}_p = \mathbf{b}$). For every scalar $c \in \mathbb{R}$:

   $$A(\mathbf{x}_p + c\mathbf{v}_h) = A\mathbf{x}_p + c(A\mathbf{v}_h) = \mathbf{b} + c(\mathbf{0}) = \mathbf{b}$$

   So every single reachable $\mathbf{b}$ has infinitely many pre-images $\mathbf{x}_p + c\mathbf{v}_h$.

3. **Why this fails for Non-Linear functions:** For non-linear maps (like $f(x) = (x-2)^2$), $f(x) = 0$ has only one solution ($x = 2$), yet $f(x) = 1$ has two ($x = 1, 3$), because $f(u) - f(w) \neq f(u - w)$. **Linearity is the exact superpower that makes $\mathbf{0}$ a universal detector for the entire space.**

Proof (Bidirectional $\iff$)

Recall: T is one-to-one $\iff$ $\big(T(\mathbf{u}) = T(\mathbf{w}) \implies \mathbf{u} = \mathbf{w}\big)$ for all $\mathbf{u}, \mathbf{w} \in V$.

1. Forward Direction ($\implies$): Assume T is one-to-one, prove $\ker(T) = \{\mathbf{0}_V\}$

1. By linearity, $T(\mathbf{0}_V) = \mathbf{0}_W$, so $\mathbf{0}_V \in \ker(T)$.
2. Let $\mathbf{v}$ be any arbitrary vector in $\ker(T)$, meaning $T(\mathbf{v}) = \mathbf{0}_W$.
3. Equating gives $T(\mathbf{v}) = T(\mathbf{0}_V)$.
4. Since T is one-to-one, $T(\mathbf{v}) = T(\mathbf{0}_V) \implies \mathbf{v} = \mathbf{0}_V$.
5. Hence, $\ker(T) = \{\mathbf{0}_V\}$.

2. Backward Direction ($\impliedby$): Assume $\ker(T) = \{\mathbf{0}_V\}$, prove T is one-to-one

1. Let $\mathbf{u}, \mathbf{w} \in V$ such that $T(\mathbf{u}) = T(\mathbf{w})$.
2. Subtracting gives $T(\mathbf{u}) - T(\mathbf{w}) = \mathbf{0}_W$.
3. By linearity, $T(\mathbf{u} - \mathbf{w}) = \mathbf{0}_W$, so $(\mathbf{u} - \mathbf{w}) \in \ker(T)$.
4. Since $\ker(T) = \{\mathbf{0}_V\}$, it follows that $\mathbf{u} - \mathbf{w} = \mathbf{0}_V \implies \mathbf{u} = \mathbf{w}$.
5. Thus, T is one-to-one.

■

Theorem: Composition of Linear Transformations is Linear & Matrix Product Rule

Statement

> Let $T_1 : U \to V$ and $T_2 : V \to W$ be linear transformations. Define the composition $(T_2 \circ T_1) : U \to W$ by $(T_2 \circ T_1)(\mathbf{u}) = T_2(T_1(\mathbf{u}))$. Then:
>
> - **(a)** $T_2 \circ T_1$ is a linear transformation.
> - **(b)** If $U = \mathbb{R}^n$, $V = \mathbb{R}^k$, and $W = \mathbb{R}^m$, the standard matrix of the composition is the matrix product $[T_2 \circ T_1] = [T_2][T_1]$.

Proofs

- **Proof of (a):** Let $\mathbf{u}, \mathbf{v} \in U$ and $c \in \mathbb{R}$.

  1. **Additivity:**

$$(T_2 \circ T_1)(\mathbf{u}+\mathbf{v}) = T_2(T_1(\mathbf{u}+\mathbf{v})) \overset{T_1 \text{ lin.}}{=} T_2(T_1(\mathbf{u}) + T_1(\mathbf{v})) \overset{T_2 \text{ lin.}}{=} T_2(T_1(\mathbf{u})) + T_2(T_1(\mathbf{v})) = (T_2 \circ T_1)(\mathbf{u}) + (T_2 \circ T_1)(\mathbf{v})$$

2. **Homogeneity:**

$$(T_2 \circ T_1)(c\mathbf{u}) = T_2(T_1(c\mathbf{u})) \overset{T_1 \text{ lin.}}{=} T_2(cT_1(\mathbf{u})) \overset{T_2 \text{ lin.}}{=} cT_2(T_1(\mathbf{u})) = c(T_2 \circ T_1)(\mathbf{u})$$

- **Proof of (b):** For any $\mathbf{x} \in \mathbb{R}^n$:

$$(T_2 \circ T_1)(\mathbf{x}) = T_2(T_1(\mathbf{x})) = [T_2]([T_1]\mathbf{x}) = \big([T_2][T_1]\big)\mathbf{x}$$

By uniqueness of the standard matrix representation, $[T_2 \circ T_1] = [T_2][T_1]$.

■

Theorem: General Rank-Nullity Theorem & Equal-Dimension Equivalence

Statement

> - **(a) Rank-Nullity Theorem for Transformations:** If $T : V \to W$ is a linear transformation from an n-dimensional vector space V into a vector space W, then:
>
> $$\text{rank}(T) + \text{nullity}(T) = \dim(R(T)) + \dim(\ker(T)) = \dim(V) = n$$
>
> - **(b) Equal-Dimension Equivalence:** If $\dim(V) = \dim(W) = n$ (finite), then T is **one-to-one** if and only if T is **onto**.

Proofs

Proof of Part (a): $\text{rank}(T) + \text{nullity}(T) = n$

Let $k = \text{nullity}(T) = \dim(\ker(T))$.

- *(Boundary cases)* If $k = n$, then $\ker(T) = V$, so $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in V \implies R(T) = \{\mathbf{0}\} \implies \text{rank}(T) = 0$, and $0 + n = n$.
- *(General case $0 \leq k < n$)* Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for $\ker(T)$ (if $k = 0$, start with the empty set). Extend this linearly independent set to a basis of all of V:

$$B = \{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{v}_1, \ldots, \mathbf{v}_{n-k}\}$$

We prove that $S = \{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_{n-k})\}$ is a **basis for** $R(T)$:

1. S **spans** $R(T)$**:** Let $\mathbf{w} \in R(T)$, so $\mathbf{w} = T(\mathbf{x})$ for some $\mathbf{x} \in V$. Write $\mathbf{x}$ in the basis B:

$$\mathbf{x} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k + d_1\mathbf{v}_1 + \cdots + d_{n-k}\mathbf{v}_{n-k}$$

Apply T and use $T(\mathbf{u}_i) = \mathbf{0}$ (since $\mathbf{u}_i \in \ker(T)$):

$$\mathbf{w} = T(\mathbf{x}) = \sum_{i=1}^{k} c_i \underbrace{T(\mathbf{u}_i)}_{\mathbf{0}} + \sum_{j=1}^{n-k} d_j T(\mathbf{v}_j) = d_1T(\mathbf{v}_1) + \cdots + d_{n-k}T(\mathbf{v}_{n-k}) \in \text{span}(S)$$

2. S **is linearly independent:** Suppose $d_1T(\mathbf{v}_1) + \cdots + d_{n-k}T(\mathbf{v}_{n-k}) = \mathbf{0}$. By linearity:

$$T(d_1\mathbf{v}_1 + \cdots + d_{n-k}\mathbf{v}_{n-k}) = \mathbf{0} \implies (d_1\mathbf{v}_1 + \cdots + d_{n-k}\mathbf{v}_{n-k}) \in \ker(T)$$

Since $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ spans $\ker(T)$, there exist scalars $c_1, \ldots, c_k$ such that:

$$d_1\mathbf{v}_1 + \cdots + d_{n-k}\mathbf{v}_{n-k} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$$

$$(-c_1)\mathbf{u}_1 + \cdots + (-c_k)\mathbf{u}_k + d_1\mathbf{v}_1 + \cdots + d_{n-k}\mathbf{v}_{n-k} = \mathbf{0}$$

Since B is a basis of V (linearly independent), every coefficient is zero, so $d_1 = \cdots = d_{n-k} = 0$.

Therefore, S is a basis for $R(T)$, giving $\text{rank}(T) = \dim(R(T)) = n - k$. Hence:

$$\text{rank}(T) + \text{nullity}(T) = (n-k) + k = n$$

Proof of Part (b): Equal Dimensions $\implies$ (T 1-to-1 $\iff$ T onto)

Assume $\dim(V) = \dim(W) = n$.

$$T \text{ is one-to-one} \iff \ker(T) = \{\mathbf{0}\} \iff \text{nullity}(T) = 0$$

By Part (a) (Rank-Nullity Theorem):

$$\text{nullity}(T) = 0 \iff \text{rank}(T) = n - 0 = n = \dim(W)$$

Since $R(T)$ is a subspace of the finite-dimensional space W, $\dim(R(T)) = \dim(W)$ if and only if $R(T) = W$, which is the exact definition of T being **onto**.

■